# Milestone 3 - Modelling and Design Work

*Vision*

The first prototype of CitySafe is designed to be a mobile-first, technically realistic concept that validates the objective of our group's vision: enabling safer urban navigation through community-driven information, and quick assistance within the community. The system reflects the vision of supporting the community, especially pedestrians, students, and tourists during navigation scenarios.

The prototype includes core functionality, allowing for a full safety interaction cycle: viewing hazards through the interactive map, submitting hazard reports, receiving notifications, and requesting assistance. These features were prioritized because they are the true nature of the app, supporting the vision of easing urban navigation and safety. Without this particular cycle, the vision would be incomplete, and the viability of the system would not be sufficiently demonstrated.

*Feature Selection with Justification*

The following section will outline which features are suggested to be fully developed, partially developed, and postponed for future updates of the software, the reasons for each suggestion will be clearly outlined.

Fully Developed in Prototype 1
These will be implemented end-to-end with working UI and backend functions.

F-1: Interactive Safety Map
This is considered a high-priority feature, as it is the primary interface through which users can interpret hazards, crowding, and alerts. Without this feature, CitySafe cannot demonstrate its capability.

F-2: Safety Alerts/Notifications
A high priority feature. Alerts are central to the heart of CitySafe's mission, appearing in scenarios where users are warned about nearby hazards. A prototype lacking this feature would not sufficiently test the system's real-time requirements.

F-4: User Reports
User reports are another essential feature to the software. Community-driven reporting supplies the app with the data needed to run as intended. Building this feature validates the feasibility of crowd-sourced reports and supports later features of credibility and verification.

F-6: Call For Help (SOS)
This is a vital feature to the core functionality of CitySafe, as it separates the software from traditional navigation apps. This feature is tied to high-priority user scenarios for assistance-based requests. It validates the community-driven nature of the app.

F-8 User Accounts & History
While user history may only be needed minimally, user accounts are a significant feature of the app. Authentication is necessary to associate reports and help requests with a user identity, and prevent the abuse of features. Full history dashboards are not essential to the core workflow, so it is suggested to be kept to a minimum.

F-9: Data Privacy
This feature is essential to the interests of our users. Security and privacy are of the utmost importance to this project, and users of the early prototype should be assured that their data will not be mishandled.

Partially Developed in Prototype 1
These features will be simplified implementations demonstrating the intended usage, but without advanced logic.

F-3: Safe Navigation
This feature will be implemented to show suggested routes that avoids flagged areas using basic logic (e.g. avoid markers), rather than fully-fledged optimised routing. The justification for this is rooted in the idea that while navigation is a priority, advanced routing introduces complexities that threaten the stability of the prototype, a simplified version of this feature is sufficient to validate user flow.

F-5: Heat Awareness
Heat awareness is suggested to be implemented through the integration of a weather API to show heat warnings and a “heat risk” indicator. This supports safety scenarios and warnings without requiring a full environmental model implementation. Building a complete engine for heat-routing would greatly expand the scope and increase dependency risk.

F-7: Display Settings
This feature will be implemented as one accessibility toggle set (for high-contrast). Accessibility is important to the goals of CitySafe, so a limited set of settings in the early prototype will demonstrate that need while keeping minimal complexities.

Postponed Development - not in Prototype 1
These features are suggested to be intentionally postponed for clear reasons.

F-11: Credibility System
While trust is important, it is not necessary to the core functionality of the app to implement a full credibility scoring system, as it requires careful anti-abuse design, and high-level testing.

F-10: Point System
This feature is suggested to be deferred to future versions of the software. This is because gamification introduces additional data, rules and abuse moderation. It is considered a low priority as it is not necessary to validate safety workflows of the app.

F-12: Tourist Recommendations

Recommendations from locals are suggested to be postponed to a later update, because it introduces complexities in content sourcing, moderation, logic and evaluation. This does not correlate directly to the app's mission, and so can be deferred.

F-13: CarPlay/Android Auto Integration
This is a future-oriented feature, and is not necessary to validate the feasibility of CitySafe. Therefore, it is better to defer it to future updates.

*Suggested Technology Stack*

Frontend

React (responsive web app)
Justification: Great response time and UI integration, reusable components such as map screen, report form etc. Reduces device/platform overhead compared to native apps while still being mobile-first.

Tailwind CSS
Justification: Accelerates UI development while maintaining consistent styling. Reduces styling overhead and allows for quick changes that reflect across the entire feature set.

Leaflet.js with OpenStreetMap
Justification: Implements interactive safety map in a cost-effective way, allowing CitySafe to be freely accessible to the community. Provides sufficient mapping functionality for prototype goals, reduces integration risk while allowing a fully functional outcome.

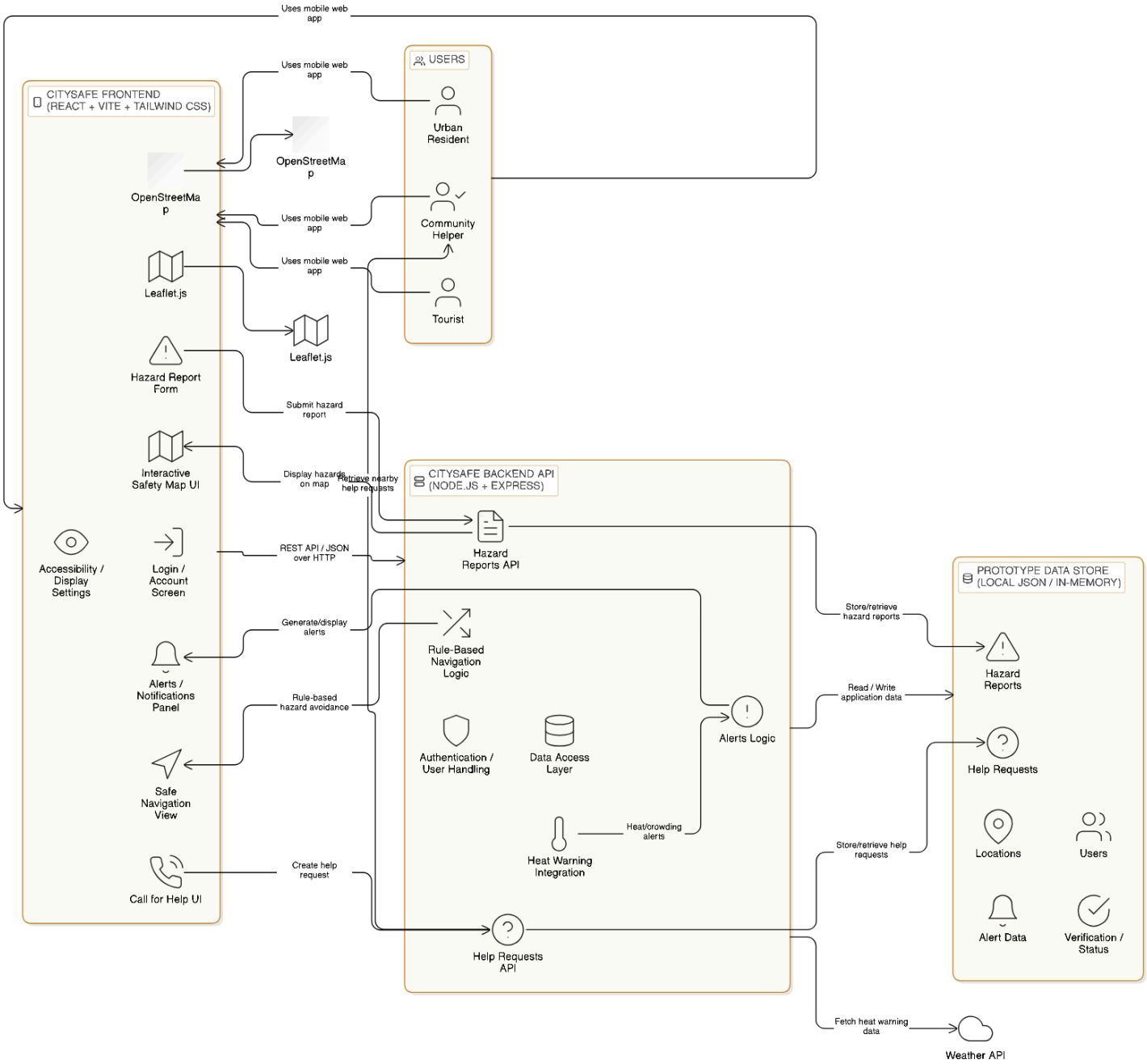
Backend

Node.js with Express (REST API)
Justification: Required to support shared state between users (reports, help requests, alerts) and to support basic validation and data handling. Node.js is lightweight, has a quick set up process, and is suited to interactions such as creating reports and broadcasting notification. Express reduces configuration overhead and supports faster delivery of the prototype.

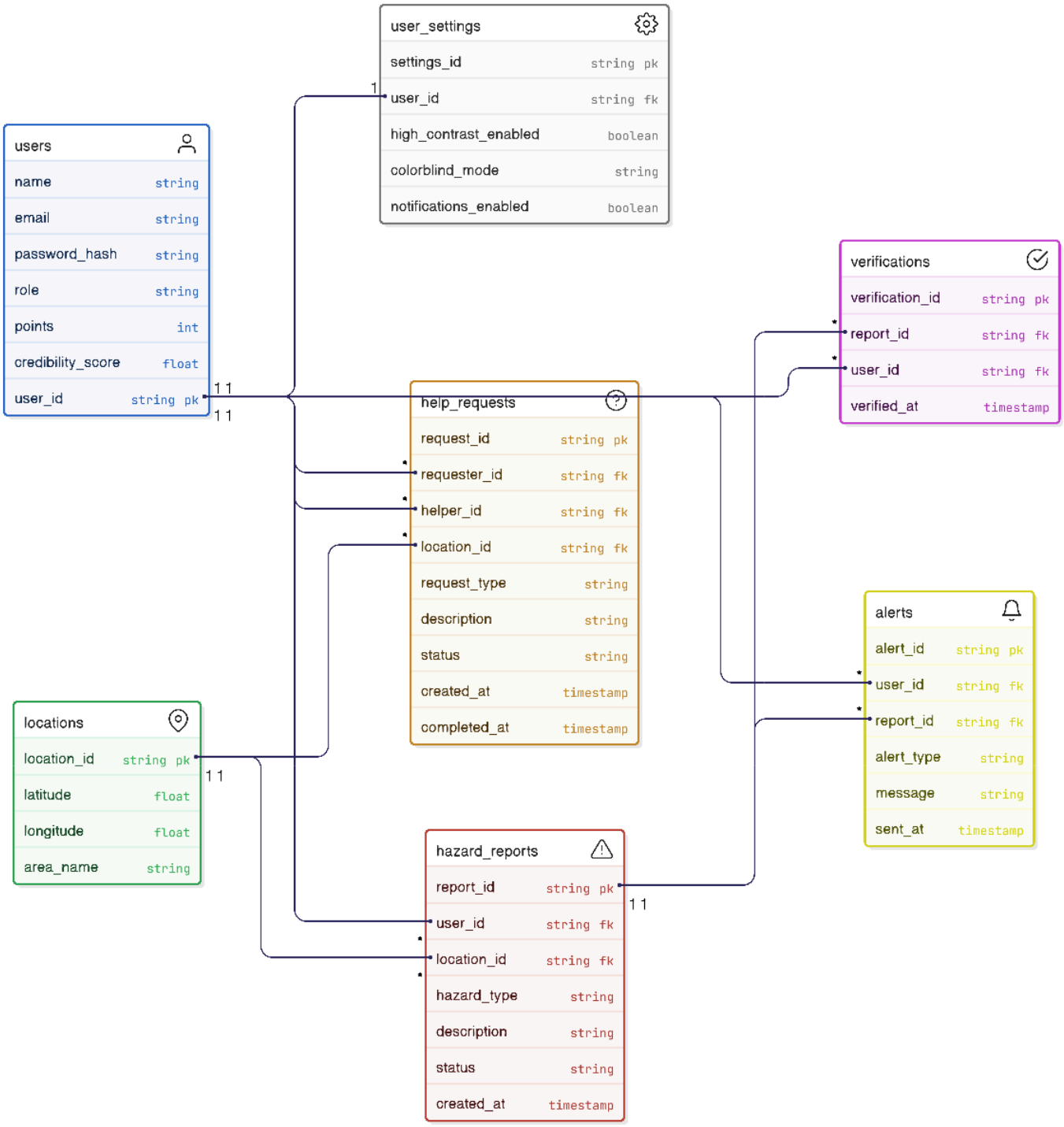
Local JSON files/in-memory storage (Mock Data)
Justification: For prototype 1, this is used instead of a fully-fledged database to reduce operational complexity and speed up development. The milestone's goal is to validate workflows and user interactions, not production-grade persistence. Therefore, a mock database will be implemented in the prototype, to allow the team to focus on UI and core behaviors.

Simple Routing Logic
Justification: The safe navigation feature will be implemented using rule-based hazard avoidance rather than complex optimization. This allows for users to be guided away from hazards, proving the feasibility of the prototype, while reducing algorithmic uncertainty and unforeseen bugs.

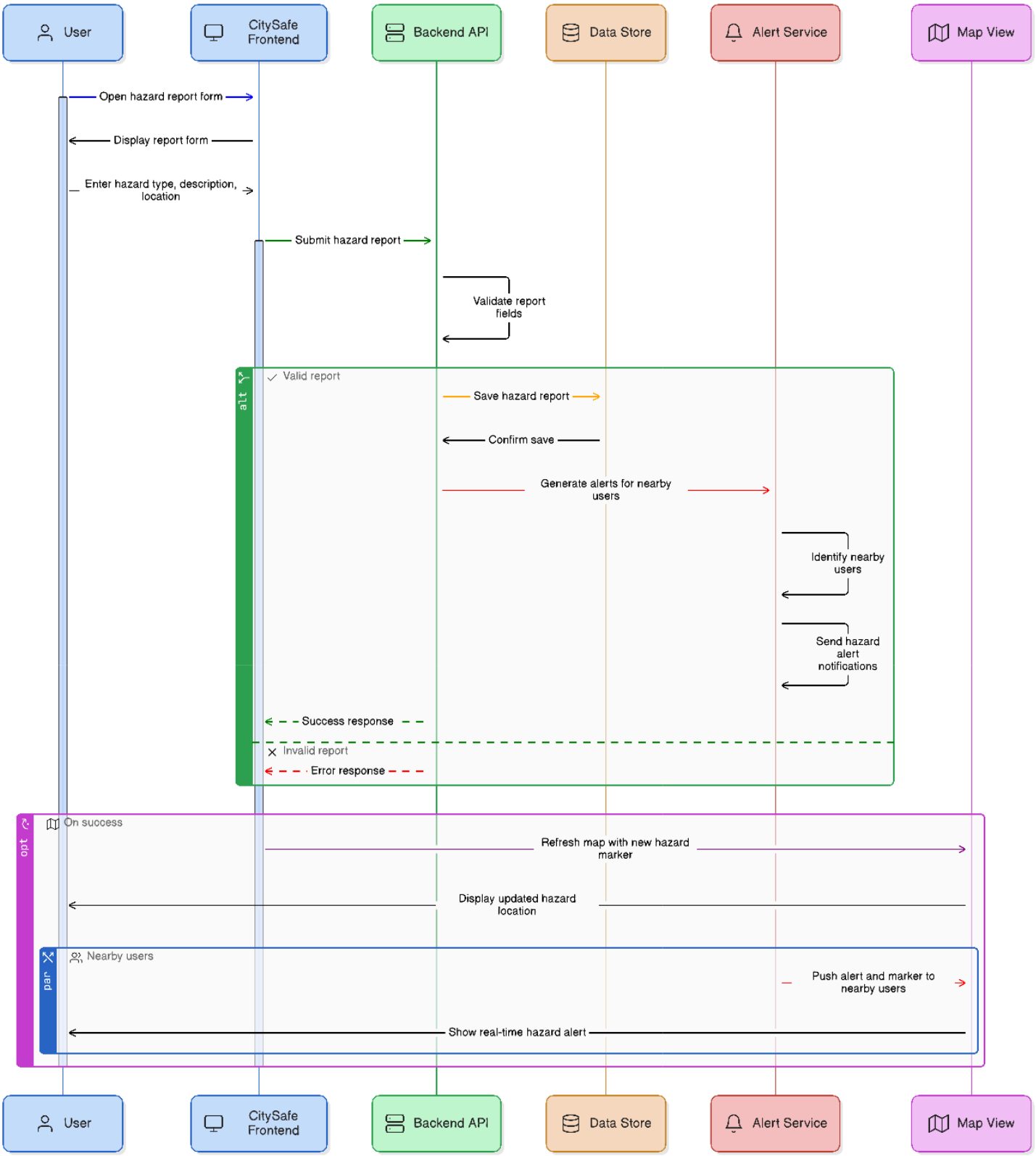
# *System Architecture Model*



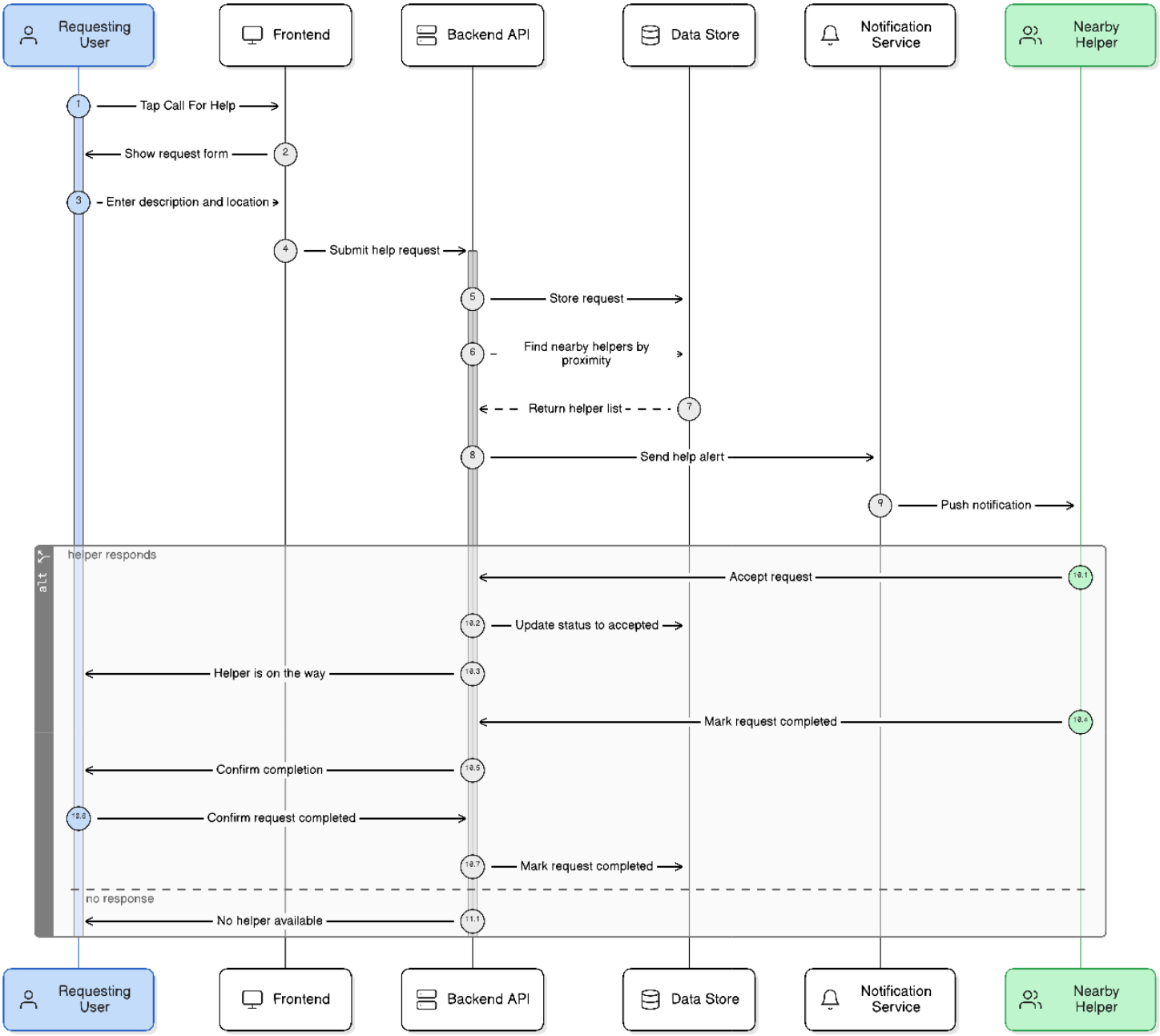
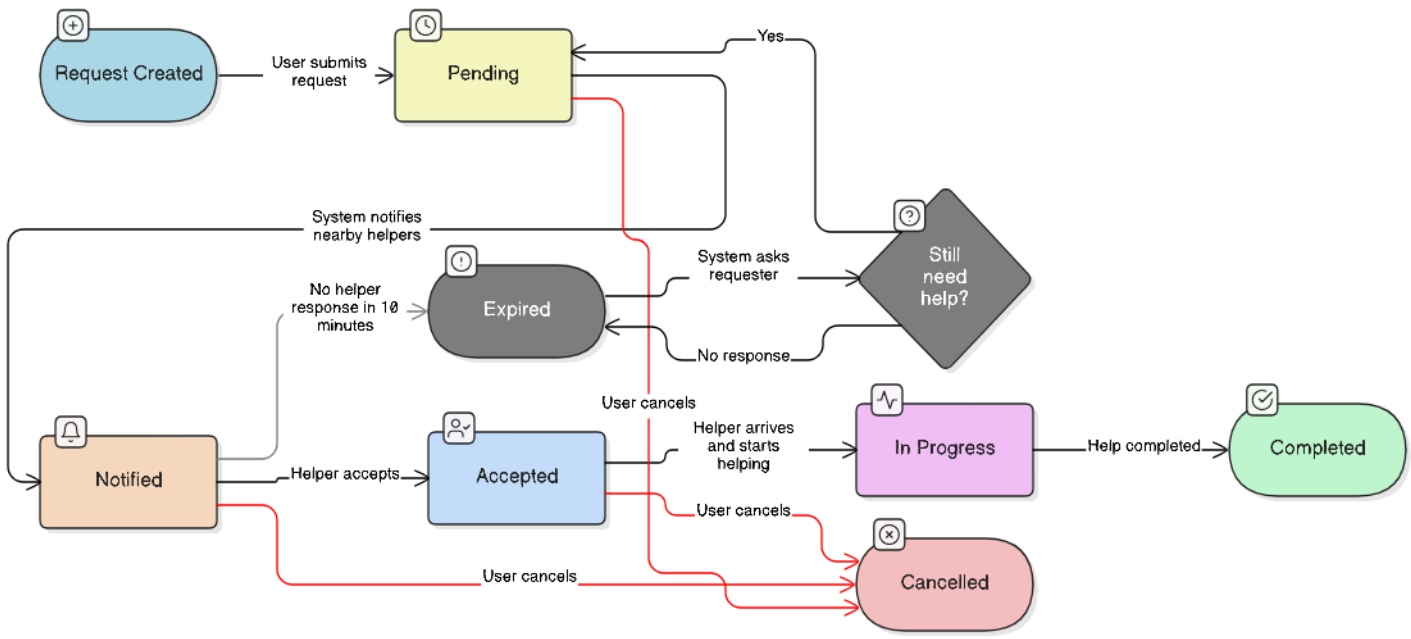
*Entity-Relationship Diagram*



*Sequence Diagram 1: Submit Hazard Report*



*Sequence Diagram 2: Call For Help*



*State Diagram: Request for Help*